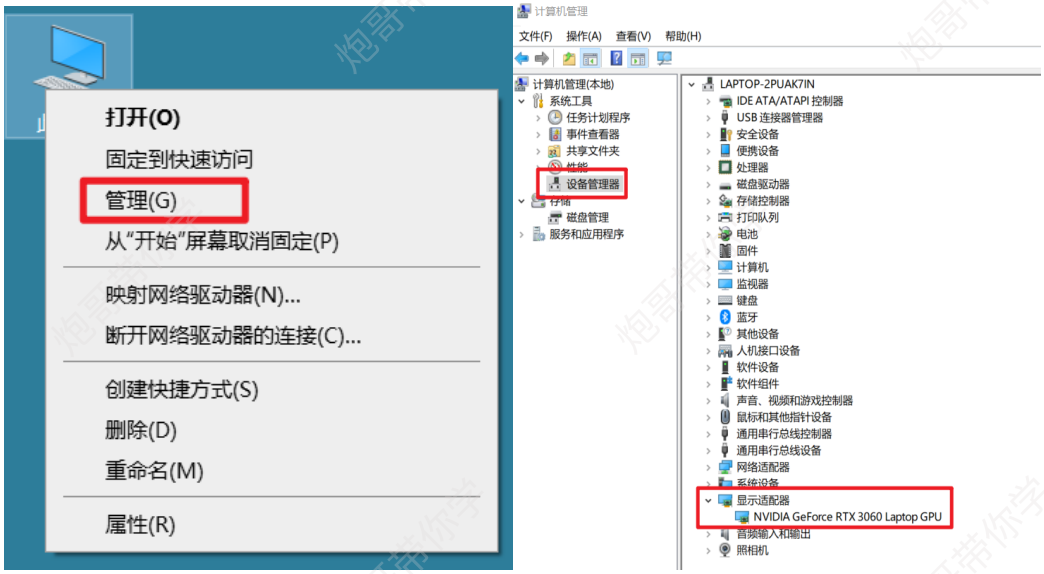
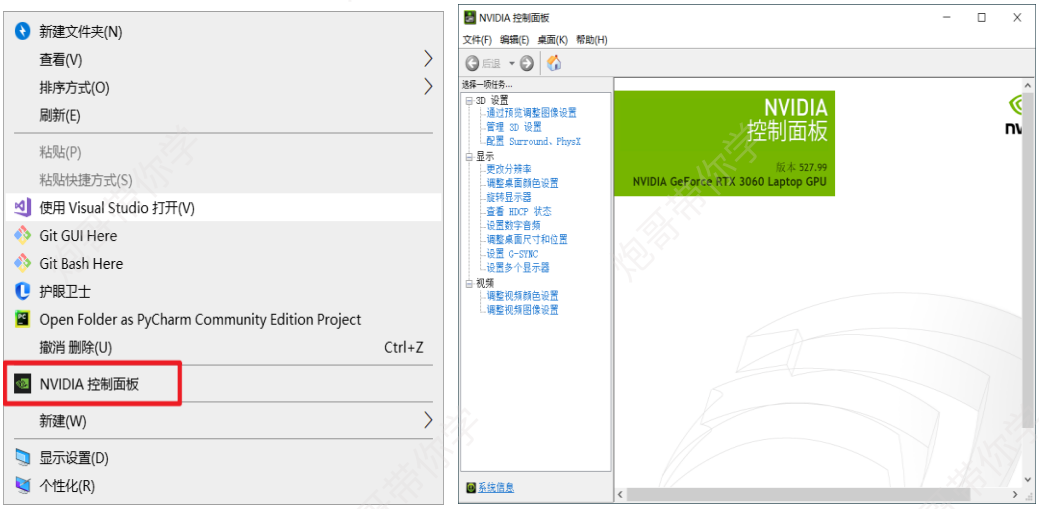
pytorch 与深度学习环境安装

NVIDIA 驱动安装与更新

按如下步骤找到自己显卡的型号，可以看到我的显卡是 3060



有显卡驱动的，可以直接在桌面右键，找到英伟达驱动控制面板打开就好了。

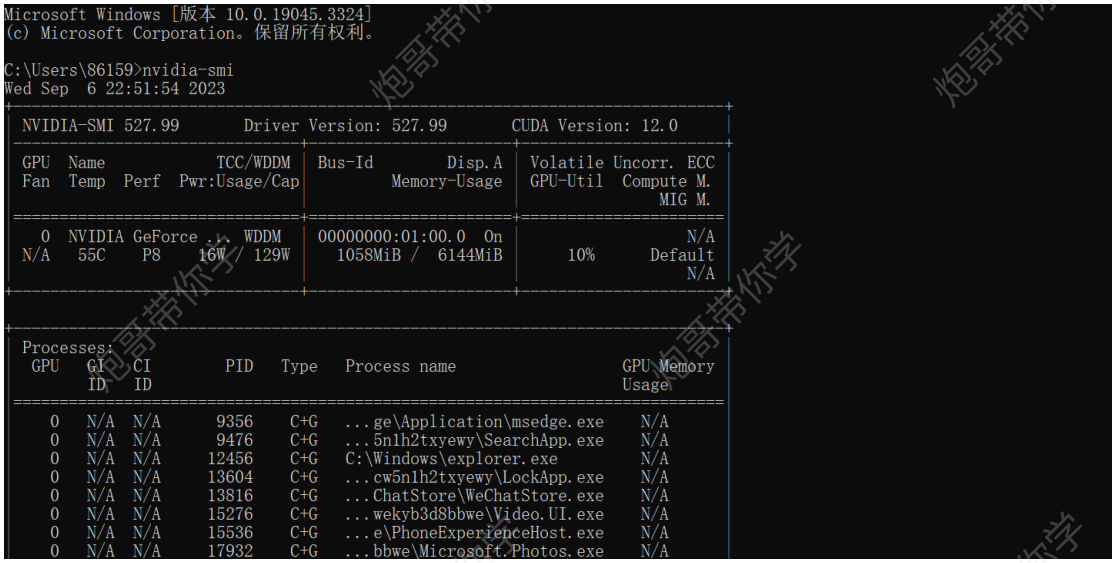


没有显卡驱动去英伟达驱动官网 https://www.nvidia.cn/Download/index.aspx?lang=cn 打开驱动官网找到适合自己设备的选项就行。



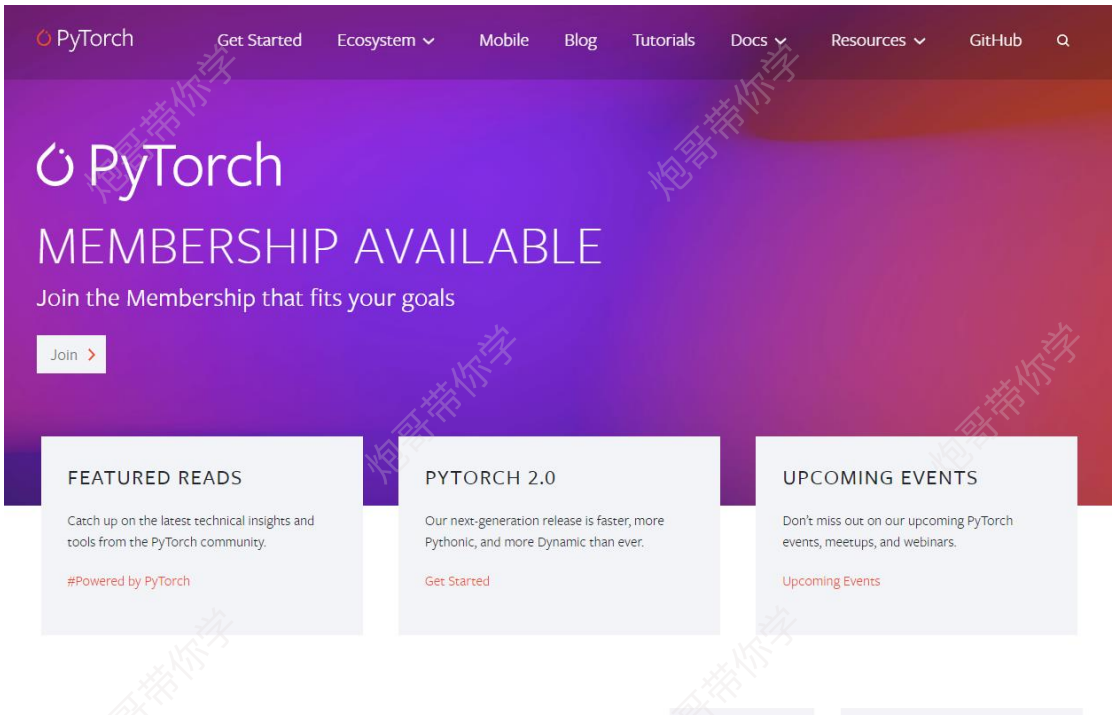
下载安装完以后，我们查看支持 cuda 驱动的支持最高版本 cuda tookit。
按下 win+R 组合键，打开 cmd 命令窗口。输入如下的命令:
nvidia-smi

得到如下的结果，可以看到我的显卡驱动支持的最高版本 cuda tookit 是 12.0，版本是向下兼容的，因此只要安装向下兼容的 cuda tookit 版本即可。

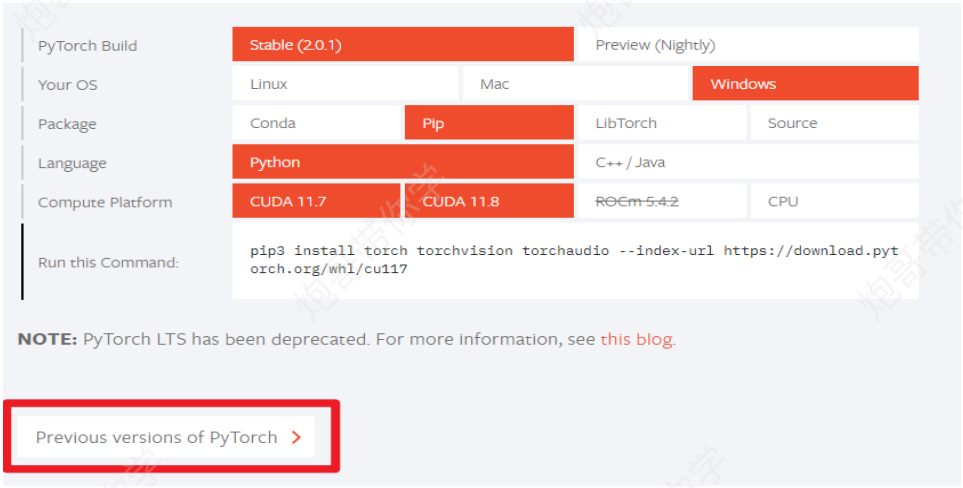


pytorch 的 GPU 深度学习环境安装

输入如下 pytorch 下载网址：https://pytorch.org/，打开的网址如下图所示（随着时间的流程官网页面也会更新）。



将页面往下滑动，出现如下界面，可以看到当前最新的版本为 2.0.1 版本，对应的 cuda tookit 有 11.7 和 11.8，安装方式有 pip 和 cuda，还有对应的电脑系统。由于我们要用的是 1.10.1 版本的 pytorch 和 cuda tookit 有 11.3，因此点图中红框的连接，找之前的版本。



不断往下滑动，找到对应的 1.9.1 版本，复制红框中的链接

v1.10.1

Conda

OSX

```
# conda
conda install pytorch==1.10.1 torchvision==0.11.2 torchaudio==0.10.1 -c pytorch
```

Linux and Windows

```
# CUDA 10.2
conda install pytorch==1.10.1 torchvision==0.11.2 torchaudio==0.10.1 cudatoolkit=10.2 -c pytorch

# CUDA 11.3
conda install pytorch==1.10.1 torchvision==0.11.2 torchaudio==0.10.1 cudatoolkit=11.3 -c pytorch -c conda-fo

# CPU Only
conda install pytorch==1.10.1 torchvision==0.11.2 torchaudio==0.10.1 cpuonly -c pytorch
```

进入 anaconda 自己创建的环境，复制上述的命令，加载对应的依赖包，输入 y，进行对应的 pytorch 和 cudatookit 安装。中间可能会因为网络的原因导致下载对应的安装包失败，就需要从新输入该命令，使其恢复下载，已经下载好的包不会在次下载，只会下载没有安装好的包。

```
(torch) C:\Users\wu>conda install pytorch==1.10.1 torchvision==0.11.2 torchaudio==0.10.1 cudatoolkit=11.3 -c pytorch -c
conda-forge
```

```
  libuv                conda-forge/win-64::libuv-1.44.2-h8ffe710_0
  libwebp-base         conda-forge/win-64::libwebp-base-1.2.4-h8ffe710_0
  libxcb               conda-forge/win-64::libxcb-1.13-hcd874cb_1004
  libzlib              conda-forge/win-64::libzlib-1.2.12-h8ffe710_2
  lz4-c                conda-forge/win-64::lz4-c-1.9.3-h8ffe710_1
  m2w64-gcc-libgfor~ conda-forge/win-64::m2w64-gcc-libgfortran-5.3.0-6
  m2w64-gcc-libs     conda-forge/win-64::m2w64-gcc-libs-5.3.0-7
  m2w64-gcc-libs-co~ conda-forge/win-64::m2w64-gcc-libs-core-5.3.0-7
  m2w64-gmp            conda-forge/win-64::m2w64-gmp-6.1.0-2
  m2w64-libwinpthre~ conda-forge/win-64::m2w64-libwinpthread-git-5.0.0.4634.697f757-2
  mkl                  conda-forge/win-64::mkl-2022.1.0-h6a75c08_874
  mkl-devel            conda-forge/win-64::mkl-devel-2022.1.0-h57928b3_875
  mkl-include          conda-forge/win-64::mkl-include-2022.1.0-h6a75c08_874
  msys2-conda-epoch    conda-forge/win-64::msys2-conda-epoch-20160418-1
  numpy                conda-forge/win-64::numpy-1.23.2-py38h223ccf5_0
  openjpeg             conda-forge/win-64::openjpeg-2.5.0-hc9384bd_1
  pillow               conda-forge/win-64::pillow-9.2.0-py38h37aa274_2
  pthread-stubs        conda-forge/win-64::pthread-stubs-0.4-hcd874cb_1001
  python_abi           conda-forge/win-64::python_abi-3.8-2_cp38
  pytorch              pytorch/win-64::pytorch-1.10.1-py3.8_cuda11.3_cudnn8_0
  pytorch-mutex        pytorch/noarch::pytorch-mutex-1.0-cuda
  tbb                  conda-forge/win-64::tbb-2021.5.0-h2d74725_1
  tk                   conda-forge/win-64::tk-8.6.12-h8ffe710_0
  torchaudio           pytorch/win-64::torchaudio-0.10.1-py38_cu113
  torchvision          pytorch/win-64::torchvision-0.11.2-py38_cu113
  typing_extensions    conda-forge/noarch::typing_extensions-4.7.1-pyha770c72_0
  xorg-libxau          conda-forge/win-64::xorg-libxau-1.0.11-hcd874cb_0
  xorg-libxdmcp        conda-forge/win-64::xorg-libxdmcp-1.1.3-hcd874cb_0
  xz                   conda-forge/win-64::xz-5.2.6-h8d14728_0
  zstd                 conda-forge/win-64::zstd-1.5.2-h6255e5f_4

The following packages will be SUPERSEDED by a higher-priority channel:

  ca-certificates    anaconda/pkgs/main::ca-certificates-2~ --> conda-forge::ca-certifi
  openssl            anaconda/pkgs/main::openssl-1.1.1w-h2~ --> conda-forge::openssl-1.


Proceed ([y]/n)? y
```

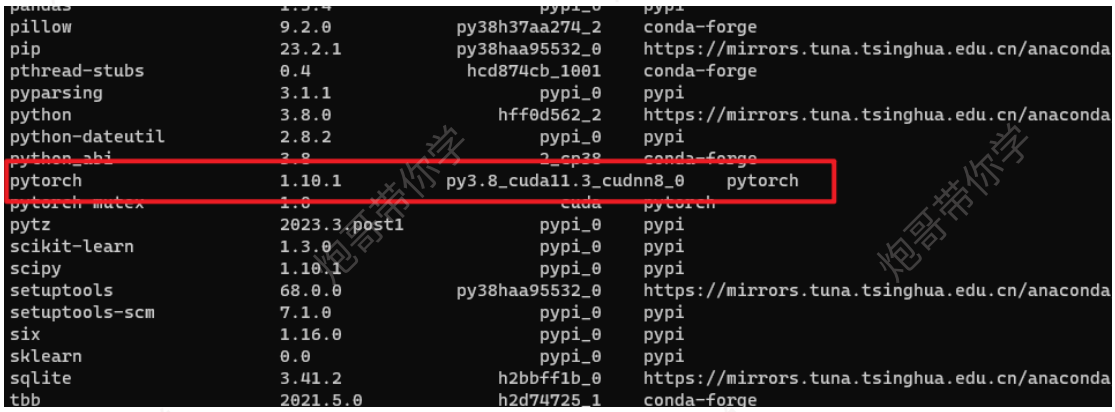
下载完成以后，输入如下的命令查看环境中的安装包：

Conda list

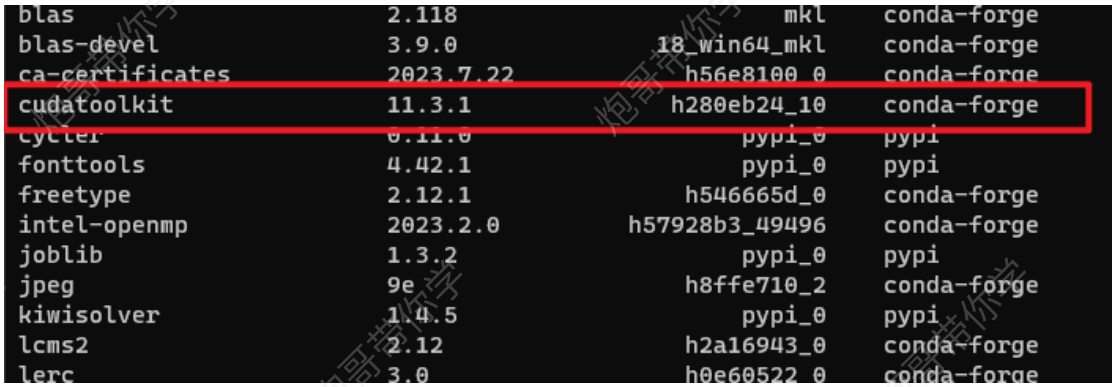
会得到该环境下的所有环境包，（该图只展示了部分）

# Name	Version	Build	Channel
blas	2.118	mkl	conda-forge
blas-devel	3.9.0	18_win64_mkl	conda-forge
ca-certificates	2023.7.22	h56e8100_0	conda-forge
cudatoolkit	11.3.1	h280eb24_10	conda-forge
cycler	0.11.0	pypi_0	pypi
fonttools	4.42.1	pypi_0	pypi
freetype	2.12.1	h546665d_0	conda-forge
intel-openmp	2023.2.0	h57928b3_49496	conda-forge
joblib	1.3.2	pypi_0	pypi
jpeg	9e	h8ffe710_2	conda-forge
kiwisolver	1.4.5	pypi_0	pypi
lcms2	2.12	h2a16943_0	conda-forge
lerc	3.0	h0e60522_0	conda-forge
libblas	3.9.0	18_win64_mkl	conda-forge
libcblas	3.9.0	18_win64_mkl	conda-forge
libdeflate	1.12	h8ffe710_0	conda-forge
liblapack	3.9.0	18_win64_mkl	conda-forge
liblapacke	3.9.0	18_win64_mkl	conda-forge
libpng	1.6.37	h1d00b33_4	conda-forge
libtiff	4.4.0	h2ed3b44_1	conda-forge
libuv	1.44.2	h8ffe710_0	conda-forge
libwebp-base	1.2.4	h8ffe710_0	conda-forge
libxcb	1.13	hcd874cb_1004	conda-forge
libzlib	1.2.12	h8ffe710_2	conda-forge
lz4-c	1.9.3	h8ffe710_1	conda-forge
m2w64-gcc-libgfortran	5.3.0	6	conda-forge

查看是否有安装好的对应版本的 pytorch 和 cudatoolkit



pillow	9.2.0	py38h37aa274_2	conda-forge
pip	23.2.1	py38haa95532_0	https://mirrors.tuna.tsinghua.edu.cn/anaconda
pthread-stubs	0.4	hcd874cb_1001	conda-forge
pyparsing	3.1.1	pypi_0	pypi
python	3.8.0	hff0d562_2	https://mirrors.tuna.tsinghua.edu.cn/anaconda
python-dateutil	2.8.2	pypi_0	pypi
python_abi	3.8	2_cp38	conda-forge
pytorch	1.10.1	py3.8_cuda11.3_cudnn8_0	pytorch
pytorch-mutex	1.0	cuda	pytorch
pytz	2023.3.post1	pypi_0	pypi
scikit-learn	1.3.0	pypi_0	pypi
scipy	1.10.1	pypi_0	pypi
setuptools	68.0.0	py38haa95532_0	https://mirrors.tuna.tsinghua.edu.cn/anaconda
setuptools-scm	7.1.0	pypi_0	pypi
six	1.16.0	pypi_0	pypi
sklearn	0.0	pypi_0	pypi
sqlite	3.41.2	h2bbff1b_0	https://mirrors.tuna.tsinghua.edu.cn/anaconda
tbb	2021.5.0	h2d74725_1	conda-forge



blas	2.118	mkl	conda-forge
blas-devel	3.9.0	18_win64_mkl	conda-forge
ca-certificates	2023.7.22	h56e8100_0	conda-forge
cudatoolkit	11.3.1	h280eb24_10	conda-forge
cycler	0.11.0	pypi_0	pypi
fonttools	4.42.1	pypi_0	pypi
freetype	2.12.1	h546665d_0	conda-forge
intel-openmp	2023.2.0	h57928b3_49496	conda-forge
joblib	1.3.2	pypi_0	pypi
jpeg	9e	h8ffe710_2	conda-forge
kiwisolver	1.4.5	pypi_0	pypi
lcms2	2.12	h2a16943_0	conda-forge
lerc	3.0	h0e60522_0	conda-forge